

THE MELAO INDEX: A NEW STANDARD FOR RISK-RETURN ANALYSIS, RESOLVING FUNDAMENTAL INCONSISTENCIES IN THE SHARPE RATIO AND RELATED METRICS

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ABSTRACT

Accurately evaluating the risk-return relationship has been one of the biggest challenges in the financial market. Although metrics such as the Sharpe (ShI), Sortino (SoI), Treynor (TI), MAR (MR), Calmar (CR), Modigliani (MRAP) and Omega (OR) ratios are widely employed, they present conceptual and quantitative errors that distort the evaluation of investments and hinder them from achieving the objective for which they were designed. This article exposes previously unidentified flaws in classical metrics, thoroughly analyzes them and presents improvements that make these indexes more accurate and efficient. In addition, introduces a new metric – the Melao Index (MeI). This not only make a comprehensive review of classic metrics, but also establishes a new standard for measuring risk-adjusted performance.

Keywords: Melao Index, Sharpe Index, MAR Ratio, Calmar Ratio, Sortino Index, Treynor Index, Modigliani Risk-Adjusted Performance, Omega Ratio, risk-adjusted performance.

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I. INTRODUCTION

The Sharpe Index (ShI) – proposed by William Sharpe in 1966 [1], Nobel Prize in Economics in 1990 [2], is widely used by banks, investment funds and other financial entities to measure the quality of an investment based on risk-adjusted performance.

The score corresponding to ShI is obtained from (1) presented in [3]:

$$ShI = \frac{R - R_0}{\sigma} \quad (1)$$

Where ShI is the Sharpe Index, R is the return on investment, R_0 is the return on the assumed risk-free *benchmark*, and σ is the standard deviation in investment returns.

Sharpe's idea is interesting and addresses a problem of great importance, but the way the formula was constructed violates some fundamental principles of Measure Theory, generating distorted and inconsistent scores, compromising its effectiveness in the process of selecting the best investments, requiring some corrections and improvements.

Several attempts have been made to improve the ShI or to replace it with a superior index, but so far there has been no success, partly because the focus of the improvements did not address the problems that are analyzed here. The two main advances had been the corrected Sharpe Index – which adds the standard deviation to the *benchmark*, assuming that there are no risk-free investments – and the Sortino Index (SoI) [4], which considers only negative volatility. These two cases will also be analyzed, but our focus will be on the problems that had not yet been considered.

To this end, an analysis of the main observed errors was conducted, and solutions were proposed, including a review of formulas to enhance consistency and accuracy. However, none of these revisions to existing indexes fully achieve the primary objective of accurately selecting optimal investment alternatives. For this reason, we introduce a new index: the Melao Index.

II. PROBLEMS WITH THE SHARPE INDEX

The core principle of ShI is that investment quality should not be assessed solely based on profitability. It is necessary to verify the risk assumed to try to achieve that profitability and determine the proportion between these two variables.

Some of the points that need to be considered are:

1. There are different ways to measure profitability, and this is not being done properly with current metrics.
2. There are different ways to measure risk, and there is some flexibility in this choice, but within each choice there are optimizations that need to be performed but are not done, resulting in distorted metrics that need to be normalized.
3. In some indexes, profitability and risk are not on the same scale, this also needs to be normalized, and both cannot be used in the same formula before this normalization.
4. Profitability and risk do not exhibit the same behavior over time, this requires the addition of a temporal parameter that continuously normalizes the scales as a function of time.
5. Large positive variations should not reduce the score in the way they do in ShI, but they should also not be completely neglected as they do in SoI.
6. There are no zero-risk *benchmarks*, *but even if there were, they would not provide the information needed to serve as a reference for the scale.*
7. The distribution of returns is not Gaussian, unlike what is assumed in ShI, and the occurrence of *outliers* can have catastrophic effects.

Some of the problems mentioned above have already been partially resolved, but the solutions adopted have generated new problems, often more serious. For example: *Calmar Ratio* (CR) [5] takes into account that the true distribution of gains and losses is not Gaussian, adopting an approach that is more sensitive to *outliers*. The problem is that the way this is done in CR produces several distortions. CR mitigates the problems mentioned in items 5 and 7, but aggravates the problem in item 4 and does not address the problems in items 1, 2 and 3, which are still present. In addition, it introduces new

problems, such as, for example, the size of the maximum *drawdown* (MDD) increases as the history becomes longer, but the formula for CR does not consider this aspect.

The SoI corrects the problem in item 5, but in doing so, it favors investments that had abnormal gains by luck, such as the Jureia fund [6] that spent years in the red until, in a lucky operation in 2013, it had huge gains, as can be seen in the graph in figure 1, raising the historical average without this abnormal movement being accounted for in the volatility by the SoI. Furthermore, it does not address the problems in items 1, 2, 3 and 7.

Among these 7 problems, the only one that has received some attention is item 6. It is known that there is no *benchmark* with zero risk, and to deal with this, a parameter has usually been added to the ShI formula – the standard deviation of the *benchmark* – producing what has become known as the volatility-adjusted Sharpe Index.

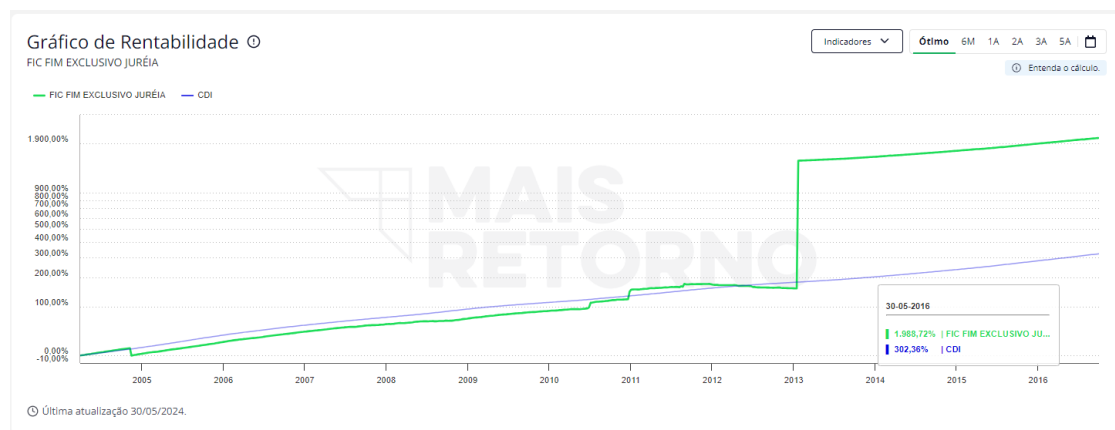


Figure 1 – Profitability graph of the Jureia fund [6].

Beyond the seven issues outlined above, the method itself has inherent flaws that require correction. Therefore, it is necessary to develop a new method to measure risk-adjusted performance that can fulfill this role in a more accurate and conceptually correct manner.

This article discusses the seven items listed above, both the parts that have already received partial solutions and those that remain unresolved, pointing out and explaining the errors present in traditional methods and how they can be corrected and improved. An innovative solution, the MeI, is also presented, exploring perspectives that had not been considered until now and can substantially change the way the quality of an investment is assessed.

The problems listed above are analyzed below, starting with the most basic ones.

III. HOW TO CORRECTLY MEASURE PROFITABILITY

When measuring the historical profitability of an investment for accounting or tax purposes, what is desired is in fact the proportion between the final and initial balance, but when the purpose is to find the parameters of the growth curve that best represent the characteristics of that investment, with the purpose of making forecasts of future performance, this is not the appropriate way to measure.

One of the problems is that if the measurement is done in this way, it becomes heavily dependent on the “choice” of the starting and ending points. Although in some cases it may not be a “choice” at all, the important thing is that the starting and ending points exert excessive weight in determining the parameters of the growth curve, so that if an investment was lucky enough to start at a “good time” and end before a “bad time”, the parameters calculated for the growth curve are overestimated. Otherwise, if the period starts at a “bad time” and ends before a “good time”, the parameters calculated for the growth curve are underestimated. In neither of these cases is it possible to make a realistic projection of future performance.

In order to predict future profitability, it is crucial to know the growth curve that best represents the historical data set. This curve should not be determined based only on the starting and ending points, ignoring the intermediate historical data. This has serious consequences, because in addition to generating great uncertainty, it does not produce reliable results about the true average growth rate.

If the monthly (or daily, or annual) balances are plotted on a graph covering a sufficiently long period so that local anomalies are diluted, one expects a curve that approximately fits an exponential function. If one replaces the variable representing the y-axis by the logarithm of this variable added to 1, the curve becomes almost a straight line and one expects to be able to make a linear regression of the points representing the balances at each moment.

On the one hand, all of this may seem trivial, as it is a common procedure in many areas, including Economics. However, if this error continues to be made, it indicates that it is not so trivial and requires a more detailed explanation of the problem, as well as the presentation of justifications as to why it needs to be this way.

The following graphs illustrate the problem of using a starting point and an ending point to determine the growth rate, comparing the lines generated by joining these points when there are small differences in the choices of which will be the starting and ending points.

A pair of points were chosen to maximize performance and another pair to minimize performance, in order to make the problem more noticeable. Next, it is shown how the arithmetic mean of the angular coefficients of these two lines is a better representation of the true growth curve than either of these two extreme lines. Finally, it is concluded that, rather than the mean of these lines based on 4 points, it is more appropriate to use all the points to perform a linear or polynomial regression, or some other appropriate function. The choice of the function used in the modeling will depend on the particularities of the historical series. The important thing is that instead of using only 2 points, the starting and the ending, as many points as are available throughout the history are used.

In general, the longer the period considered, the more the regression line tends to resemble the line connecting the initial point to the final point, especially the slope of the line tends to be increasingly similar. However, for relatively short periods, of a few decades or a few years, the line produced by connecting the initial balance with the final balance is strongly dependent on the choices of the initial and final dates, which interfere in the measurement of the slope of this line, remembering that the slope of this line corresponds to the growth rate.

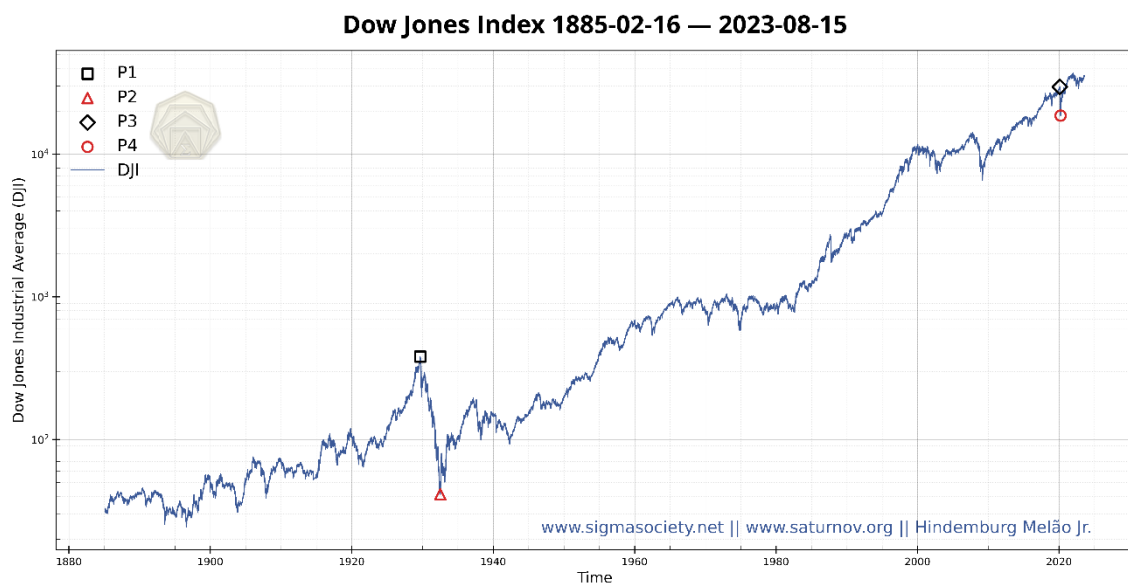


Figure 2 - Cumulative return of the Dow Jones index from 1885 to 2023 on a logarithmic scale.

Figure 2 shows the DJI with four points (P1, P2, P3 and P4) chosen to exemplify this problem.

Between September 3, 1929 (P1) and March 23, 2020 (P4), growth was from 381.17 to 18,591.93, almost 5,000%.

But if a slightly different period is chosen, from July 8, 1932 (P2) to February 12, 2020 (P3), the growth goes from 41.22 to 29,551.42, more than 70,000%.

Over a 90 year history, a change of just three years in the choice of starting and ending points caused a huge difference in the cumulative performance measure. Furthermore, the profitability in the shorter interval was greater than the profitability in the longer interval.

Sensitivity to start and end points makes the traditional method inadequate for representing average growth reliably.

In order to have a more representative measure of correct performance in the period and more useful for predicting future results, it would be necessary to consider not only the starting and ending points, but also the set of all intermediate points, that is, it would be necessary to perform a linear regression, or some other regression, depending on the purpose.

This proposal cannot be accepted without first briefly discussing the pros and cons. The first conceptual objection that arises when considering using linear regression is that the final point (final balance) is incorporating all intermediate points (all accumulated gains and losses), so there is redundancy in reusing all intermediate points. But this interpretation is incorrect, because although the final point is the accumulated result of all intermediate points, the individual information about the balance sheet positions at each moment of the time series is lost when only the two extreme points are considered, and this implies the suppression of most of the data, which could even indicate the shape of the regression curve over time.

It is worth noting that the morphology of the curve is valuable information that helps to estimate whether the results are the result of luck or an effective strategy. If the curve is more closely aligned to a straight line (using logarithms of the variable on the y-axis), it indicates a greater probability that an effective strategy is being used when it comes to a fund.

In figure 3, this situation can be seen more clearly.

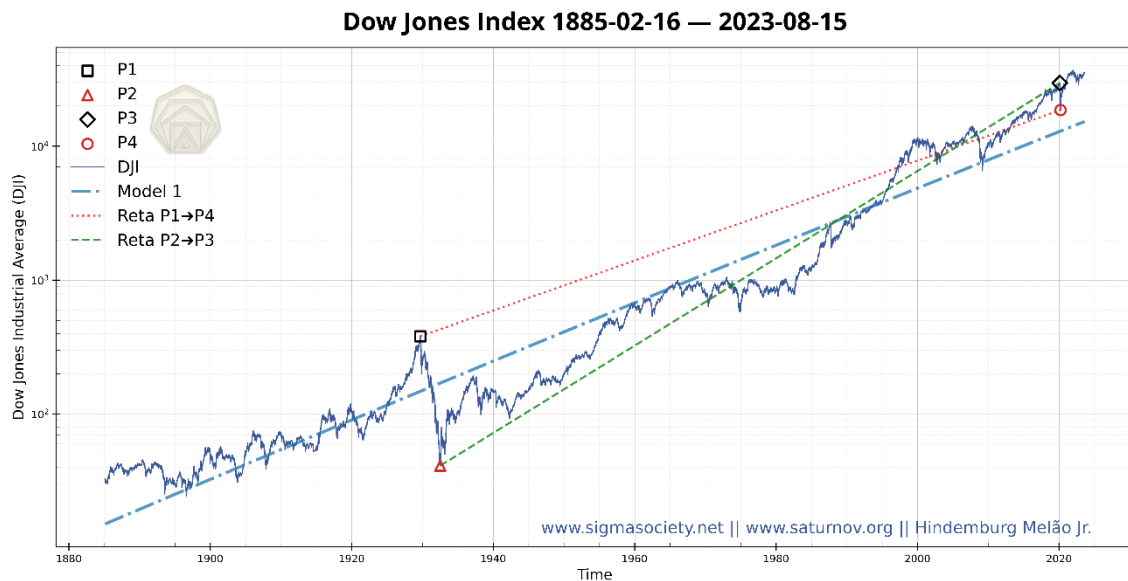


Figure 3 - Cumulative return of the Dow Jones index from 1885 to 2023 on a logarithmic scale, straight lines joining points P1—P4 and P2—P3 and linear regression of the balance sheets.

The dotted line (· ·) described in the legend as “Straight line P1 → P4” joins points P1 and P4, which are the maximum before the 1929 crisis and the minimum after the falls caused by the 2020 pandemic. Because it starts at a peak and ends at a trough, the slope of this line is very undervalued, indicating a slower growth rate than the true one. In contrast, the dashed line (— —) described in the legend as “Straight line P2 → P3” joins points P2 and P3, which are the minimum after the 1929 crisis and the maximum before the falls caused by the pandemic. In this second case, the slope of the line is overvalued, indicating growth above the correct level.

Comparing these two lines, we can conclude that it would be better to use the average of both, as a better representation of the true growth rate. And from this, we immediately generalize that, instead of using the average of both, based on only 4 extreme points, the ideal would be the average of all the points that are recorded.

The dotted-dashed line (· — · — · —) described in the caption as “Model 1” is a linear regression based on the entire set of balance sheets. As can be seen, this line is the one that offers the best representation of the set of results, with the most accurate estimate of the average profitability of this investment.

Alternatively, regressions can be made as close to the data as necessary. In Figure 4, the curve represented by “Model 1” (dotted-dashed) is a third-degree polynomial regression and provides some additional information. In this case, it indicates accelerated growth,

since the curve is concave. The “Model 2” curve uses a sophisticated adjustment method [7], whose characteristics are analyzed in more detail in [8] and [9].

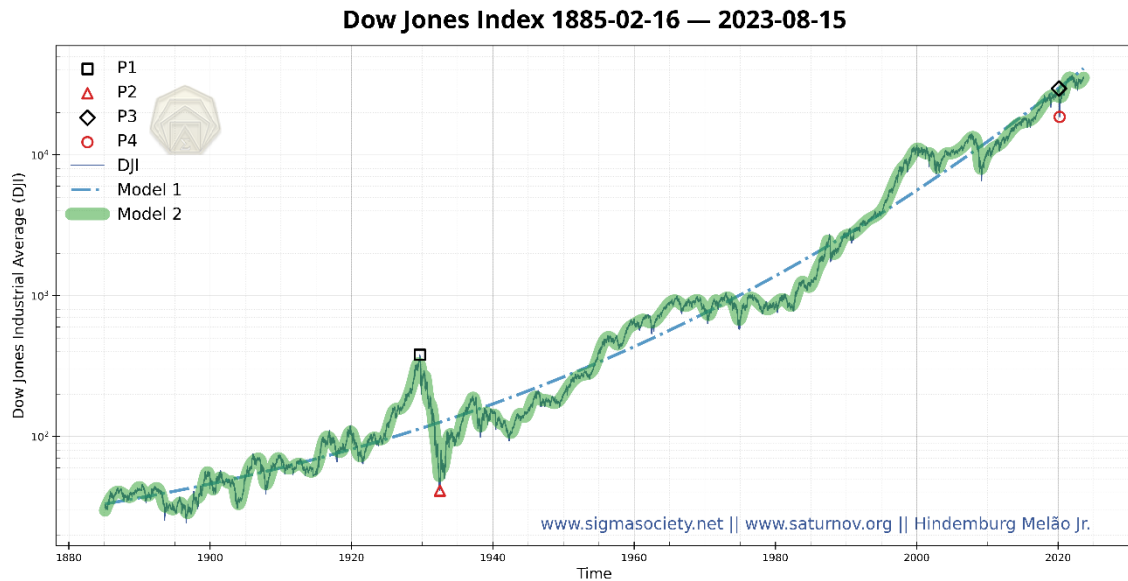


Figure 4 - Cumulative return of the Dow Jones index with model 1 (dashed dot) and model 2 (green).

Model 1 is useful for assessing whether long-term behavior indicates an acceleration or deceleration in the growth rate. Model 2 indicates local or short-term variations, but is overly sensitive to random fluctuations. It can better indicate what is happening at a given moment, with precise interpolations, but it does not help in future projections, as it does not capture global characteristics of the long-term history. Model 1 may be able to reasonably predict what the likely future of this curve will be in the coming years or even decades, but model 2 is not useful for making predictions, as it is overly adjusted to small fluctuations, which can be interpreted as *overfitting*. Model 2 would be useful for more precise data interpolations (for filling in missing data, for example), but not for extrapolations. It can also be used to smooth the curve, in some indicator calculations, among other purposes.

It is important to note that, despite there being a history of acceleration, it cannot be assumed that such behavior will continue. There is not enough evidence to support this hypothesis, because much of the effect that produces this curvature in comparison to the regression line comes from the 1929 crisis, which is closer to the beginning of the history than to the end. In a history of 138 years, there was only one event responsible for this behavior of the curve, and the fact that this event was close to the beginning generated the “illusion” of acceleration in the growth rate. If there were another recent similar event – with similar amplitude and duration – the slope would be corrected to almost a straight

line. Therefore, the projection of future profitability based on the regression line may be more accurate than that based on the curve of the degree 3 polynomial. There are other more sophisticated ways of making such a projection, but this is beyond the scope of the article.

For the purposes of this analysis, it is sufficient to examine the case of linear regression, which provides a reliable estimate of the historical behavior and future projections of a given investment. By comparing the slope of the regression line with those of the other two lines, it becomes clear that regression is the most accurate way to obtain realistic projections for the future profitability of an investment, as well as to represent the historical average.

Therefore, the first change to be implemented in the risk-adjusted profitability measure proposed in this article is to replace the conventional method of measuring profitability with the one described above. Depending on the purpose of the analysis, when performing the regression, the most recent balance sheets should be given greater weight, because they include a greater number of months and reflect more recent scenarios. In addition, several other details can be configured according to the particularities of each situation, including the regression method, the regression function, the weighting criteria, among others.

For legal and accounting purposes, it may be necessary to consider the beginning and end as determined by established standards, but for scientific purposes of assessing the quality of an investment, this alternative is actually superior in several respects, and the advantages are particularly notable in situations where there is a large concentration of profits in a short period, usually resulting from luck, while the rest of the history shows much lower performance. In such cases, the regression curve will naturally penalize this heterogeneity, reducing the probability that cases of gains due to luck will be confused with cases of gains produced by the use of efficient strategies.

Depending on the function used in the modeling, the curve may no longer have monotonic growth, making it necessary to investigate whether monotonicity is a necessary property and, if so, adopt a function that imposes this characteristic, if an adjustment with these properties is tolerable. Other improvements may involve robust or semi-robust regression methods, minimally sensitive to large short-term fluctuations; weightings may be made by assigning greater weight to the most recent data; there may be “intelligent” models that attempt to detect seasonal oscillation patterns and reproduce this property, among

other possibilities. But, in general terms, the simple use of a linear regression already substantially improves this metric.

IV. DISCUSSION ON “RISK-FREE INVESTMENT”

When R_0 is used in (1), that is, the return of the supposedly risk-free *benchmark*, a “risk-free” *benchmark* is not actually being used. The *Melao Index* (MeI) proposed in this work also corrects this flaw, but this aspect does not constitute an innovation.

Instead of using a risk-free investment, the appropriate approach would be to utilize the inflation index, as inflation “distorts” (determines) the scale on which the variable is measured. Employing a benchmark such as interest rates, rather than inflation, does not adequately correct this distortion, as it conflates real price fluctuations in goods and services with the profit-driven motives of financial institutions. Therefore, although interest rates are commonly employed for such purposes, they are not the most suitable choice.

Inflation continuously shifts the scale’s zero point at a variable rate over time, leading to multiple distortions. Beyond altering the zero point, it also results in irregularly spaced intervals. This is notable in periods of hyperinflation, such as in Hungary after the Second World War, in Germany in 1923 or in the USSR in 1989, and with some frequency in countries in South America, Asia and Africa, including Brazil in the early 1990s. To restore scale uniformity and maintain consistent intervals, inflation must be accounted for.

The use of a benchmark (such as an interest rate) is not inherently incorrect, but its function in this formula should be to assess ‘opportunity cost’ rather than “*risk-adjusted performance*”. And for this, it would not need to be a risk-free *benchmark*; it would only need to compare the profitability of the investment with that of the *benchmark* and compare the risk of the investment with that of the *benchmark*, both adjusted for inflation. For more details, see the analysis in Appendix A.

If the *benchmark* and the investment are not within the same jurisdiction – meaning they are subject to different inflation rates – two legitimate approaches arise, though they yield different results:

1. Convert one country's currency into the other's and apply the same inflation rate to both.
2. Use each country's currency and respective inflation rate.

Ideally, the variation in the price of 1 gram of gold measured in reais between January 1, 2000 and January 1, 2020, corrected for inflation in Brazil (INPC), should be equal to the variation in the price of 1 gram of gold measured in dollars between January 1, 2000 and January 1, 2020, corrected for inflation in the US (CPI). In other words, the devaluation of the real in relation to the dollar should be determined by the relationship between inflation in Brazil and inflation in the US, but this is not exactly what happens in the real world, and because this is not the case, inconsistencies arise.

Since the dollar is the strongest currency, it can be treated as a “preferred inertial reference”, so to speak, so when there are other monetary units involved, the most correct procedure (in the current economic scenario, as well as in the last 50 years, but subject to changes) may be to convert any other currency into dollars, before performing the calculations, and apply the US inflation (CPI), that is, the procedure described in 1. The procedure described in 2 would not be “wrong”, but it is preferable to standardize in this way.

It is also important to note that using inflation instead of a *benchmark* standardizes the score. As proposed by Sharpe and used to do, there is an uncontrolled freedom that allows each fund to use any *benchmark* it wants, making it difficult to compare funds presented in different sources. When all funds appear in a *ranking* on the same platform, the problem is mitigated, because the standardization is resolved, but other distortions that favor some low-quality funds and harm others of better quality remain.

It is worth noting that even if all funds adopted the same *benchmark*, it would not solve the problem, because it is not the subtraction of an arbitrary *benchmark* that repositions the zero of the scale at the correct zero point. To correctly adjust the position of the zero, it is necessary to subtract inflation, and the choice of which country to consider inflation, and which of the inflation indices, is as described above, there are some complexities, but the best standardization should seek to choose the most stable monetary unit, with the most “comprehensive” and best-weighted inflation in the respective country.

Standardization of the period considered is another important element, because a fund that calculates the ShI annually will have, on average, a result 3.46 times greater than if it calculated its ShI monthly. This happens because profitability grows, on average, approximately in the same proportion as the size of this interval, but the standard deviation grows approximately in proportion to the square root of this interval. Therefore, using a period of twelve months, the result will be, on average, $\sqrt{12}$ times greater. The formula does not specify the interval that should be used and there is no consensus on this. In fact, it does not make much difference whether it is used monthly or annually, as long as the same one is always used for all investments compared, but in practice, sometimes the ShI is seen annually, other times the monthly one.

For statistical reasons, it would be slightly better to standardize monthly or even daily usage, because it would increase the number of elements (periods) in the sample, reducing uncertainty in the measurement.

The correction proposals for ShI and SoI presented in this article are analyzed below:

1. *Melao—Sharpe Index (MShI)*

$$MShI = \frac{\ln(1+R) - \ln(1+i) - \ln(1+R_0)}{\ln(1+\sigma) - \ln(1+\sigma_0)} \quad (5)$$

1. *Melao—Sortino Index (MSoI)*

$$MSoI = \frac{\ln(1+R) - \ln(1+i) - \ln(1+R_0)}{\ln(1+\bar{\sigma}) - \ln(1+\bar{\sigma}_0)} \quad (6)$$

The formulas for *MShI* and *MSoI* are the same, but the terms σ and σ_0 have different meanings. In *MShI* these terms respectively represent the standard deviation in investment returns and the standard deviation in *benchmark returns*, while in *MSoI* represent the standard deviation in negative investment returns and the standard deviation in negative *benchmark returns*.

Along the same lines, equivalent corrections determine the *Melao—Treynor Index (MTI)*, *Melao—Modigliani Index (MMI)*, *Melao—Stirling Index (MStI)*, and other indexes, simply by adjusting the scales and adopting the other corrective procedures for these other indexes. The important thing that needs to be understood is the reasons why the corrections are necessary.

Using the natural (neperian) logarithms of the values, rather than the values themselves, is essential to correct for scale distortions.

When two variables, x and y , increase in arithmetic progression (AP), the relation x/y can be used to establish a proportion that preserves their equivalence. However, if x and y increase in geometric progression (GP), to preserve equivalence it becomes necessary to use the relation $\ln(1+x)/\ln(1+y)$. This was not done correctly in ShI or SoI, which generates distortions. The formulas proposed above correct this.

In short:

- **Growth in arithmetic progression (AP):** When x and y grow linearly (in AP), the simple ratio x/y maintains a direct and proportional relationship. Since both grow in fixed increments, this relationship is an intuitive way to establish proportional equivalence between the values of x and y .
- **Growth in geometric progression (GP):** In a GP, where variables grow exponentially, a simple ratio x/y does not preserve proportional equivalence between x and y , since exponential growth alters the relative rate of change between values. In such cases, logarithmic transformation, such as $\ln(1+x)/\ln(1+y)$, becomes essential to linearize exponential growth and avoid distortions in the results.

This logarithmic transformation approach is useful in modeling where exponential growth needs to be compared in a linearized way, as occurs in areas such as Economics, Physics and Biology. The use of the logarithm helps to “delinearize” the function, enabling direct comparisons between variables that grow at multiplicative rates.

These improved versions of MShI and MSoI resolve some inconsistencies, correct some conceptual and quantitative problems, but some fundamental problems still remain. For example, the distribution of returns is not well represented by a normal (Gaussian) distribution, so the standard deviation is not an appropriate measure of dispersion. One way to deal with this would be to test different distributions to find one that fits the experimental data, and use the parameter that represents the dispersion of this distribution instead of the standard deviation. But this would produce new difficulties and distortions, because the distribution that best represents the returns of a given investment is usually not the same as the one that best represents the distribution of the returns of the *benchmark*, nor is it the same for other investments, so using different measures of dispersion would result in distortions.

There are different ways to deal with this problem. One solution that could preserve isometry in the profitability and dispersion measures would be to ignore the morphology of the distribution and consider only the MDD. This has some positive and some negative effects. One of the positive effects is that it would solve the problem exposed in the paragraph above, because using the MDD there would be no dependence on the shape of the distribution, but only on the maximum negative amplitude. One of the negative effects is that the MDD is based on only a single extreme value, instead of considering the mean of a set of values. This makes the uncertainty in the MDD greater than the uncertainty in the standard deviation.

The use of MDD brings other advantages, the most important of which is the more efficient prediction of *outliers*. This is an extremely important difference.

In certain problems, it is desirable to use robust estimators to filter out the effects produced by *outliers*, to prevent them from contaminating the “normal” behavior of the variable of interest. Robust central tendency estimators, such as Tukey’s *biweight* [10] or Andrews’ waves [11], are more effective than the mean in determining the correct central region where the mass of data is concentrated, even if the sample is contaminated, because they filter out the interference of possible *outliers* that are very far from the central tendency. Semi-robust estimators, such as Huber’s M [12], instead of adopting a criterion to “cut” *outliers*, try to assign less weight to *outliers* according to their distance from the central tendency, and in practice this usually works worse than simply cutting, as Tukey’s *biweight* does, especially when the *outliers* are measurement errors or contamination.

Here we can cite the Hertzsprung—Russell diagram [13], which, although it deals with Astronomy, illustrates this effect very well. We can see that there is the main sequence that determines the evolution of typical stars, which align almost on a straight line, and there are clusters far from the main sequence, which could not be included in the regression used for typical stars, because this would not contribute in any way to improving the regression fit and would also unjustifiably alter the slope of this line, by treating elements that are not part of the group as if they were part of it. Figure 5, taken from [14], demonstrates this effect more clearly.

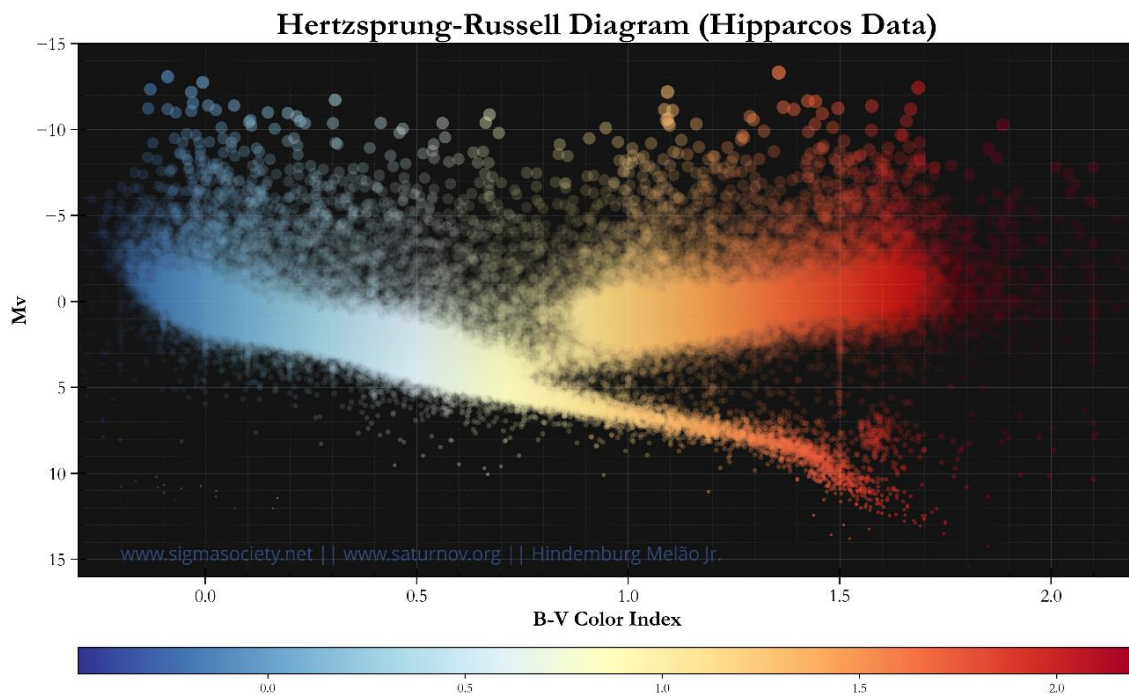


Figure 5 - Hertzsprung—Russell (HR) diagram showing the relationship between the absolute magnitude (M_v) and the BV color index of several stars, exemplifying how a set of elements can present two or more clusters that require independent curves for the data to be correctly represented and modeled.

The diagram displays a range of star colours, from blue (higher temperatures) on the left to red (lower temperatures) on the right, highlighting the different spectral classes along the main sequence. This plot is essential for illustrating stellar distribution and evolution, with hotter, more luminous stars located in the upper left of the diagram and cooler, less luminous stars in the lower right.

That's why in Astronomy it is very common to use *biweights*. Tukey 's model for this purpose, because many times the *outlier* is just a measurement error, or there may be a very pronounced contamination in a sample, or a sporadic disturbance that will not repeat itself, and even if it does "repeat", there is no reason to assume that it is similar to the previous *outlier* or has any relationship with the previous *outlier*, so that including the information from the previous *outlier* does not contribute to improving the model; on the contrary, it tends to worsen it. But when working with risk management in determining the size of the maximum risk, one should not filter out the *outliers*. On the contrary, it is important to develop methods that are sensitive to *outliers* and capable of modeling and predicting their occurrence, because it does not matter if the event is rare, what matters is that events of this type can break a portfolio if there is no effective prevention against such occurrence.

V. CONTEXTUALIZING THE TREATMENT OF OUTLIERS AND ITS PRACTICAL IMPLICATIONS WITH THE TRAGEDY IN FUKUSHIMA, ROBERT WADLOW AND INTELLIGENCE QUOTIENT

A tragic example of inadequate treatment of *outliers* in risk management occurred when the walls were built around the Fukushima nuclear power plant [20] in Japan in 1967. Initially, the plant was planned to be built about thirty meters above sea level, but for various reasons the plan and location were redesigned, and it was eventually built ten meters above sea level, surrounded by a six-meter wall. Based on the mean and standard deviation of recorded tsunamis, this seemed to be sufficient to avoid problems. The design also provided for earthquakes with an intensity of up to 8.7 on the Richter scale. One problem is that tsunami heights are not well represented by a normal distribution. Another problem is that the measurements are inaccurate. Furthermore, these heights are not constant or equal along the entire coastline. There are records of tsunamis dating back to 2100 BC [21], but even the most recent records do not include much detail and are not sufficiently precise and accurate. But the main problems are associated with the inadequacy of the hypothesis of normality of the statistical distribution, which leads to underestimations of the risks of occurrence of very large events.

In 1991, the first yellow alert was triggered [22], with the flooding of the basement of one of the reactors, indicating that measures should be taken, as it became clear that the risks had been underestimated. In 2000, an audit was carried out [20] which highlighted the need to improve tsunami prevention, but the results of this audit were only released many years later, under the claim that *“announcing information about uncertain risks would create anxiety”* [20]. Of course, all risks are uncertain, that is why they are risks, and generally the political authorities who make decisions on these matters do not have a good understanding of science and statistics.

In 2007, the head of Fukushima Daiichi, Masao Yoshida, revisited the issue, promoting a study to assess the safety of the facilities and, once again, the risk of flooding was raised. However, these risks were not taken very seriously, because tsunamis ten meters high in that region were considered very unlikely, and tsunamis of fifteen meters even more unlikely, with no similar records in recent history.

This approach, however, is seriously mistaken. When there are sufficient reasons to assume that a variable is normally distributed in the interval between -3σ and $+3\sigma$,

estimates can be made based on this, within the considered interval. But estimates cannot be indiscriminately extrapolated to values far outside this interval. In this case, there were several flaws in the criteria adopted to calculate the sufficient height of the wall. Even if they had used a Gumbel [15] or Pareto [16] distribution, currently more common to model the distribution of tsunami heights, these are still not sufficient to predict the height that an *outlier can reach*. However, if the necessary measures had been taken in 2007, the plant would most likely have been much better prepared for what would come in the following years.

To better understand this situation, let us consider a variable that has a distribution well represented by a normal distribution in the interval between -2σ and $+2\sigma$, and continues to be reasonably well represented by a normal distribution up to the interval of -3σ and $+3\sigma$, but not outside this interval. This variable is the height of people.

Height is a variable that, in the world population, has an average of 1.707 meters and a standard deviation of 0.071 meters. A normal distribution describes very well how height is distributed in the range between 1.55 meters and 1.85 meters, and still works reasonably well between 1.50 meters and 1.90 meters. However, above 2 meters and especially above 2.1 meters, the parameters of a normal distribution can no longer be used to predict the frequency with which very tall or very short people will appear.

Based on the parameters of the distribution, if the mean were exactly 1.70 meters with a standard deviation of 0.07 meters and the distribution were exactly represented by a Gaussian, the probability of there being a person taller than 2.70 meters would be 1 in 7,444,884,031,001,222,299,597,057,995,698,831,658,743,496,115.

Since the world population is “only” 8,200,000,000, and the number of people ever born throughout human history is about 250,000,000,000 (about half of whom reached adulthood), then the probability of there being someone 2.70 meters tall would be many orders of magnitude lower than the probability of winning all the lotteries by playing just one ticket in each. Therefore, if these hypotheses were correct, it would be quite reasonable to assume that there simply are not and have not been people taller than 2.70 meters. The problem is that these hypotheses are not correct, because although the distribution of heights is well represented by a Gaussian within the range from -3σ to $+3\sigma$, it stops being well represented outside that range.

Indeed, Robert Pershing Wadlow (1918—1940) [17] measured 2.72 meters tall and had a wingspan of 2.74 meters, and his height was documented by doctors and anthropometrists, who attested to the authenticity of this information, unlike other alleged giants, whose legendary heights were never formally verified, and in cases where they were measured after death, their true heights were found to be much smaller than claimed. But in Wadlow 's case, he was measured while he was alive by several independent experts, and they confirmed that he reached 2.72 meters at the age of 22.



Figure 6 - Robert Pershing Wadlow (1918—1940), the tallest man ever recorded.

The probability of a person reaching 2.72 meters (assuming the mean is 1.70 meters, the standard deviation is 0.07 meters, and the distribution is normal) is 1 in 468,526,582,122,104,923,744,964,501,402,877,149,451,172,569,274.

However, in a sample of less than 25,000,000,000 people born from the 20th century onwards, such a case occurred.

When we consider a university with twenty thousand students, and the theoretical model tells us that 16% of these students should be over 1.77 meters tall, we see that the prediction is accurate and the number of students observed over 1.77 meters actually matches what was predicted. For heights over 1.84 meters, the prediction is 2.3% and the result also matches the prediction. Up to about 1.90 meters, the predictions are still good, but the true shape of the curve becomes more anomalous, no longer being well represented by the theoretical model. Therefore, when trying to estimate the probability

of someone reaching 2.5 meters or 2.7 meters, it is not appropriate to base it on the hypothesis that height continues to have a normal distribution outside the range of -3σ and $+3\sigma$.

Figure 7 shows a comparison of the theoretical distribution, based on the normality hypothesis, and the “true” curve, adjusted to the empirical data.

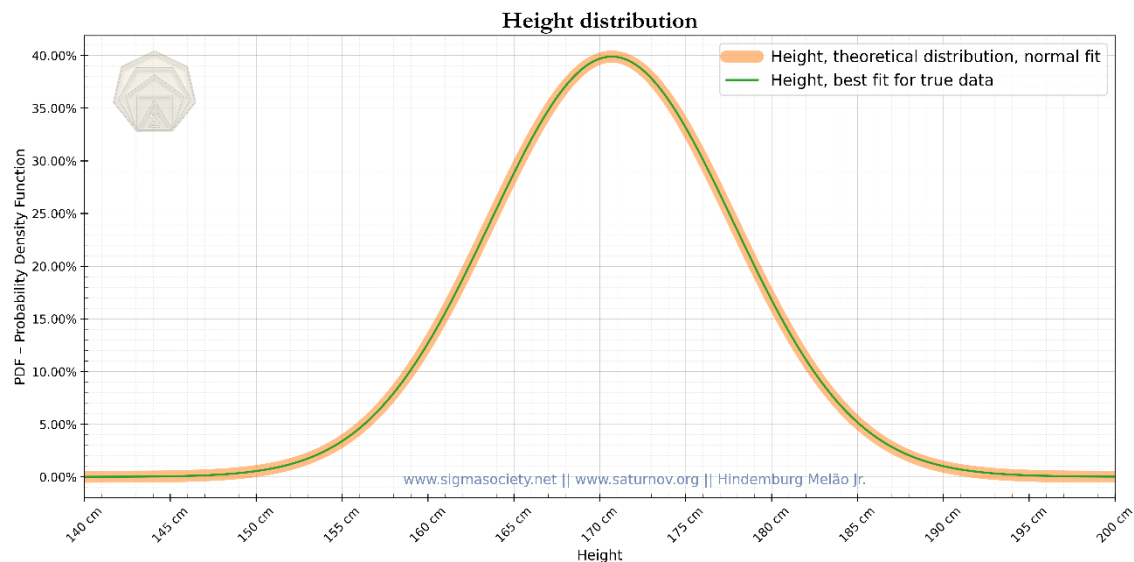


Figure 7 – Probability distribution of heights.

We can see that the normality hypothesis is an excellent representation for more than 99.9% of cases, with the two curves practically overlapping while the heights are between 1.40 meters and 2.00 meters, to the point that no difference is visually perceived. But the difference is there. We cannot see it for two reasons:

1. The disparities between the two curves only begin to grow rapidly from 1.95 meters onwards.
2. The incidence frequencies above 1.90 meters are very small, so the curve almost touches zero on the abscissa axis, making it difficult to see the differences in this region of the graph.

Figure 8, with a logarithmic scale on the y-axis, makes it easier to visualize this difference between the curves.

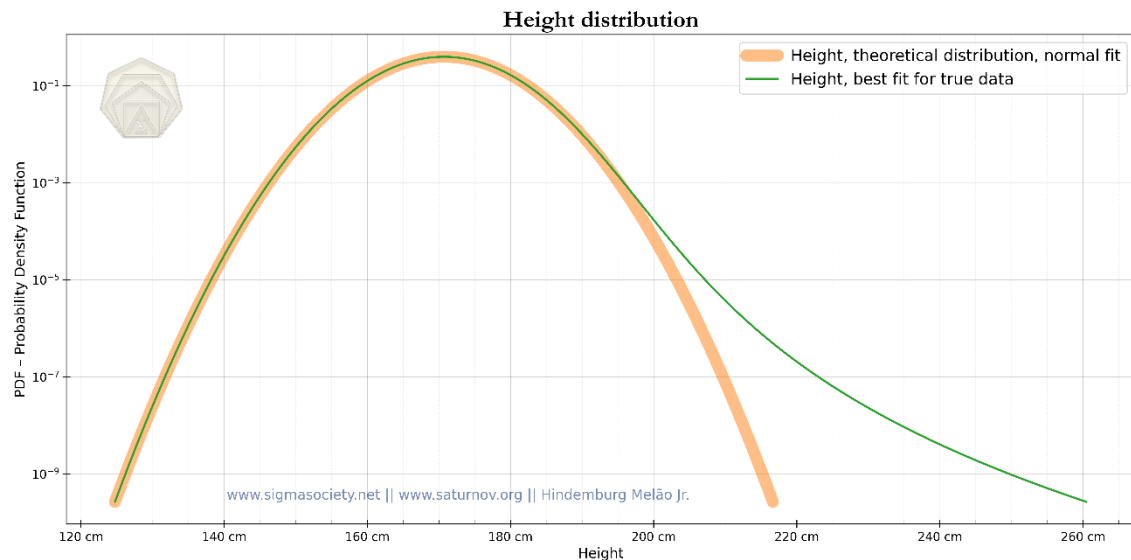


Figure 8 – Probability distribution of heights with logarithmic scale on the y-axis.

Now it is possible to clearly see what happens for heights above 1.90 meters and we can see that the two curves are practically the same for heights below 1.90 meters, which is why the normality hypothesis works very well to describe the percentages of incidences of different heights in the population, as people above 1.95 meters are very rare, and the difference between the theoretical and true distribution only starts to become noticeable above 2.00 meters.

When considering people who are much taller, the frequency with which they are observed is much higher than that predicted based on the assumption of normality, but since these cases are very rare, this error in the model goes unnoticed. If the expectation based on the model is that there is one person in every 1 billion with a height above x , but the number observed is one in 1 million, this observed number is a thousand times greater than the predicted number, a gigantic error in the model. However, since these errors only affect one in every 1 million people, such errors are almost not noticed, and there is a risk of not realizing the need to correct the model.

A similar problem occurs with the distribution of IQs (intelligence quotients), but in the case of IQ the error in the model is greater and becomes noticeable for values closer to the mean. In addition, the corrected model is morphologically more different from the traditional theoretical model, making the difference easier to visualize, without the need to use a logarithmic scale on the y-axis. [23] [24]

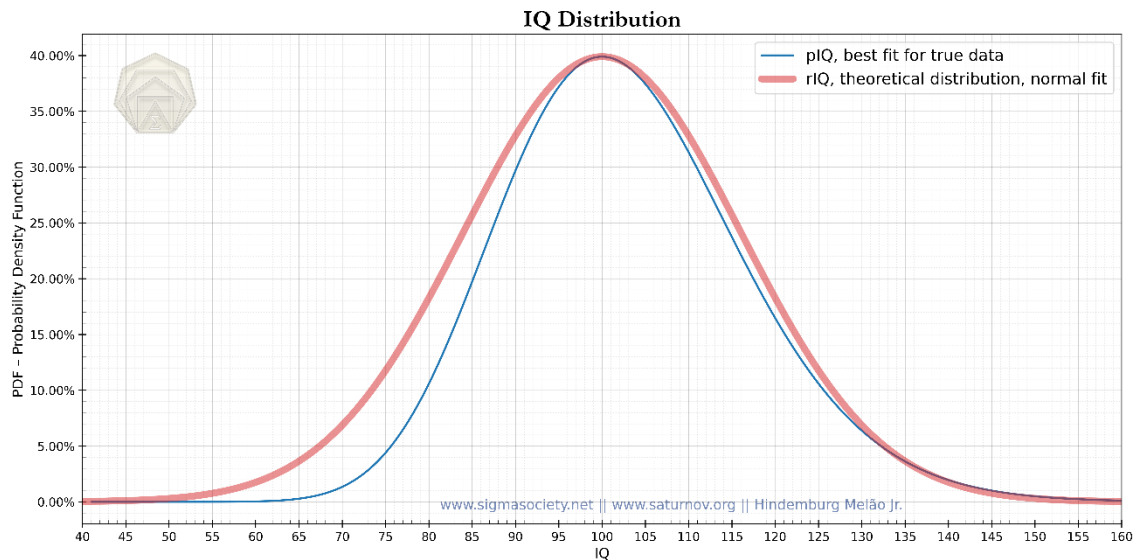


Figure 9 - Probability distribution of IQs (intelligence quotients).

In Figure 9, we can see the **blue curve (pIQ, real distribution)** that best represents the real distribution of IQs. It shows a slight asymmetry with a peak slightly shifted to the right, indicating a slightly higher concentration of individuals with above-average IQs, in addition to a denser tail on the right. The **red curve (rIQ, theoretical distribution)** attempts to represent the distribution of IQs using a Gaussian, symmetric and mesokurtic curve. Both have a mean of 100 and a standard deviation of 16, but their shapes are very different, showing different concentrations in different locations of the distribution. Both model relatively well the observed numbers of occurrences of IQs between 70 and 130, but adopting the normality assumption underestimates by many orders of magnitude the number of occurrences of very high IQs, not reflecting what is empirically observed.

To better visualize on the graph what happens at the ends of the curve, it is useful to also use a logarithmic scale.

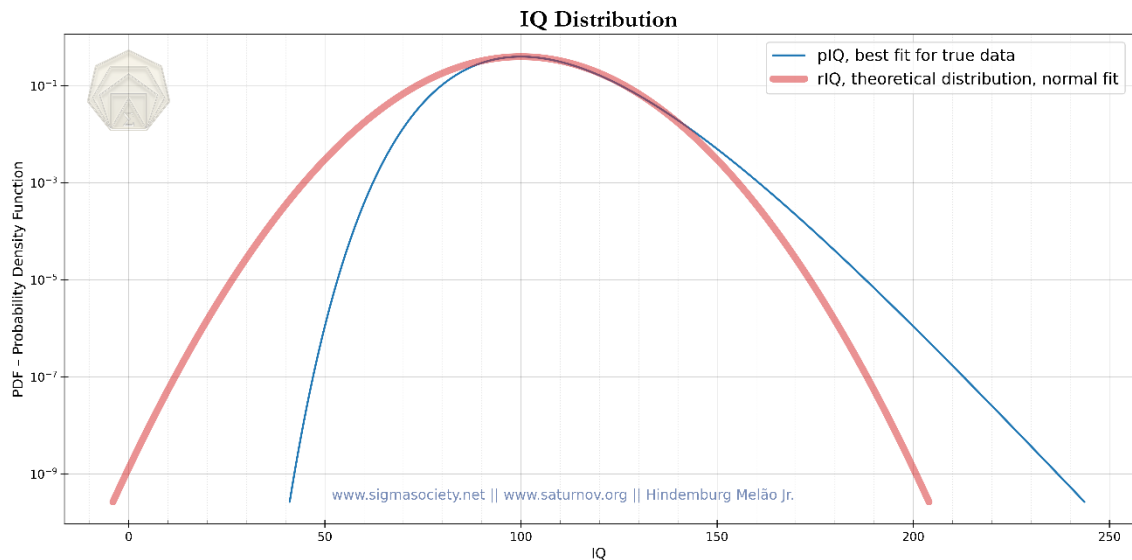


Figure 10 - Probability distribution of IQs (intelligence quotients) with logarithmic scale on the y-axis.

In Figure 10 we can observe more clearly what was described in the case of Figure 9, with a much sharper decline in the rate of occurrences for a normal curve for higher IQ values. Furthermore, the asymmetry is also more notable in the curve that most closely adheres to the real distribution of scores.

With a logarithmic scale, we can better see that, although the difference between the curves decreases above 150 IQ, the proportion between them continues to increase. We can also observe that there is an asymptotic limit at around 40, which reflects one of the consequences of using multiple-choice tests, which, below a certain point, cease to represent IQ and start to represent the probability of a random correct answer. This property of the model was a “choice” to maintain adherence to the empirical data, but a limit of zero, for example, or any other point could have been chosen, or no asymptotic limit could have been left, which would be more in line with IQ itself, rather than the result measured by the tests.

The important knowledge to be drawn from these examples is that in both the case of heights and IQs, when a model based on the normal distribution is adopted, the percentage of people predicted to have a height above a certain level, or an IQ above a certain level, is very different from the actual percentage observed, because the model underestimates the actual number of people who should have a certain height or a certain IQ, and this error exceeds the empirically observed numbers by many orders of magnitude.

But that’s not all. The vast majority of models based on the normal distribution underestimate the true number of observed occurrences by more than three or four

standard deviations, and since the normal distribution is used in almost everything in the world around us – in industry, science, medicine, politics, education – the consequences of this type of error can be catastrophic.

Defects in car parts, toxicity rates in fish, blood sugar levels in obese people, and practically everything that has an approximately normal distribution within the range of -2σ to $+2\sigma$, when samples are examined in sufficient numbers to be able to accurately count occurrences with three or four or more standard deviations above the mean, it is found that the curve tumbles and the true distribution observed ceases to be similar to the normal distribution.

When the number of elements in the sample is small and the number of samples is small, the distribution may appear less adherent to a Gaussian distribution, including in situations where the “true” distribution is Gaussian, but in these cases, it is enough to increase the number of samples for the data to approach a normal distribution. However, as in the examples above, this does not happen.

When the true distribution is not normal, the normality assumption is still generally satisfactory for small samples and for small numbers of samples, leading to incorrect interpretations. However, as more data is collected, with a sufficiently large number of samples, each of which is sufficiently numerous, the model can be improved and verified that the normal distribution often does not offer an accurate and faithful representation, even though it may seem to be an excellent representation within the range of -2σ to $+2\sigma$.

Therefore, depending on the importance of the study and the severity of the consequences if the model is wrong, it is essential to carry out a rigorous and accurate review, applying the necessary adjustments so that the corrected model is capable of making realistic predictions about the frequencies that can be expected for events well above or well below the average.

In the case of Fukushima, in 2011 a tsunami that reached almost 14 meters above sea level flooded the site, causing one of the worst nuclear accidents in human history. After the accident occurred, a review of the facts revealed that a few people had warned about the risks, but since the overwhelming majority of so-called experts did not have a good understanding of the subject, the majority opinion prevailed, no safety measures were taken, and the disaster ended up happening.

For this reason, using MDD rather than standard deviation is preferable. For theoretical models and academic research, standard deviation can provide convenient data to study, but when dealing with real money management, or with the safety and lives of real people, as in the case of Fukushima, it is very important to adopt the most correct procedures so that risk control can predict events accurately and realistically.

There are a few more details that need to be considered:

Benchmark performance also needs to be measured including the effect of inflation.

2. Compound gains grow in PG, not in PA. Therefore, instead of (1), (2) is more appropriate.

When R , R_0 , σ , σ_0 are small, the distortion from treating the case as PA is small, which makes it harder to “feel” the problem empirically, but the distortion is there, and ideally should be removed.

Although this correction is capable of eliminating some flaws and mitigating others, there are several problems intrinsic to the model that could not be eliminated without reformulating the entire model. For example, when $R < i$ or $R_0 < i$, inconsistencies occur. Therefore, instead of patching up the ShI and others in which there are inherent chronic problems, the ideal is to create an appropriate index from scratch. With these findings, the MeI seeks to correct these intrinsic problems of the performance indexes.

This does not invalidate the other corrected indexes – mainly MSoI and MShI – which can still serve specific purposes well.

VI. ASYMMETRY BETWEEN GAINS AND LOSSES

One of the issues with ShI is that it treats positive and negative volatility equally. If a fund experiences several growth spurts without any significant drawdowns, its volatility increases, unfairly penalizing the ShI score and distorting the metric. However, having large growth spurts should not be penalized in the same way that several deep drawdowns are penalized.

SoI fixes this problem by considering only negative volatility, i.e., declines from the last high-water mark. As a result, in most cases SoI ends up performing the function it was designed for better than ShI, but there are some exceptions.

Such exceptions occur because correctly assessing the quality of an investment is not easy. For example: a very large asymmetry between positive and negative volatilities may indicate lucky gains in an isolated operation. This is clearly seen in the Jureia fund, which illustrates the type of situation in which the ShI can work better than the SoI. In 2016, the Brazilian fund that best represented this type of anomaly was the Jureia fund, which is why it was chosen as an example. The Jureia fund had a smooth growth curve, around 1% per month between April 2004 and January 2013. Suddenly, from one business day to the next (01/18/2013 to 01/21/2013), it had an abnormal growth of 460% in 1 day, forming a “giraffe neck on the chart”. In such cases, the ShI is more sensitive to detecting lucky gains and penalizing them accordingly, which is why, in the case of the Jureia fund, the ShI works better than the SoI, but these cases are rare. In most “normal” cases, the SoI is superior to the ShI because it does not penalize positive volatility.

Although ShI detects some undesirable symptoms that go unnoticed by SoI, this does not mean that ShI is the most suitable tool for this purpose, or even that it is a good tool for it. There are appropriate metrics for this purpose, which do not involve ShI. The comment above, with the example of the Jureia fund, is only to show that SoI is not always better than ShI.

The important thing to be clear about is that the ShI characteristic of indifferently penalizing positive and negative volatility is a mistake (although it can be useful in special cases). It is also important to emphasize that the SoI does not adequately correct this mistake, in addition to generating some undesirable side effects. But weighing the pros and cons, for “normal” funds the SoI is superior to the ShI in most typical situations.

The Jureia fund is a textbook example in which the anomaly is very striking, but in most cases the anomaly is much more subtle, harder to detect, and consequently it is more difficult to decide in which cases ShI works better and in which cases SoI works better. In the Jureia fund, the entire history is reasonably homogeneous, with a single exception that forms the giraffe neck, but in most cases this is not what is observed. The most common is that there are upward movements that are, on average, slightly larger than downward movements, or vice versa. Furthermore, without a smooth growth curve, it is more difficult to determine the correct position of the curve from which the distances should be measured.

Therefore, SoI offers a more effective solution than ShI in most situations, but it still presents flaws that produce distortions in the scores and lead to errors in rankings and

choices. This will be analyzed later. In these cases, MeI emerges as a more efficient solution.

VII. *OUTLIERS AND FAT TAILS*

Another problem present in both ShI and SoI is that they both assume standard deviation is an adequate measure of profit dispersion. However, in practice, the tails of these distributions are dense, as explained in section III, which means that the true probability of reaching a certain level of loss is higher than what would be predicted based on the hypothesis assumed when using ShI or SoI. This is an extremely negative and dangerous issue because the way risk is measured in ShI and SoI underestimates the actual risk.

Sharpe's idea is reasonable from a theoretical standpoint and can be applied to most funds most of the time without significant issues. The issue is that it fails precisely with the best funds, those that are three or more standard deviations above the mean, which are exactly the ones where detecting weaknesses correctly is most important, as they offer the best risk/reward ratio and are the top candidates for sophisticated investors.

This problem has been addressed with unsatisfactory metrics such as *MAR Ratio* (MR), *Calmar Ratio* (CR), *Sterling Ratio* (SR) [18], among others, each of which presents flaws more serious than the errors they attempt to correct.

The formula for Calmar Ratio is $CR = \frac{CAGR}{MDD}$. For MR, the formula is the same, but it only considers the last three years.

The basic idea of CR is correct, i.e., using historical MDD instead of standard deviation as a measure of risk, because what matters most is not the average loss, but rather the maximum loss, however the standard deviation does not provide appropriate data to make reliable projections about maximum loss.

A straightforward example: suppose an investment (**X**) whose worst months of each year reach between negative 20% and negative 60%, over thirty years, with an average negative volatility of 35%. Now suppose another investment (**Y**), which reaches a maximum of between negative 1% and 2% in twenty-nine of the thirty years in its history, and only once reaches a negative 99%, with an average negative volatility of 4%. Which of these investments involves greater risk? The one with negative volatility of 35% and maximum loss of 60%? Or the one with negative volatility of 4% and maximum loss of 99%?

It is quite evident and intuitive that the highest risk is in the investment with a maximum loss of 99%. It would be even more obvious if we considered losses closer to 100%, such as 99.5% or 99.9%. Therefore, the historical MDD is a more effective parameter for determining the level of risk. That is why CR is superior to ShI and SoI in this regard, but fails in other aspects.

To begin with, one of the mistakes of CR and MR lies in how the MDD is used to generate the score. The MDD is on a scale with a range between 0 and -1, while profitability is on a scale with a range between 0 and ∞ . You cannot simply divide the profitability value by the MDD value without first normalizing the variables, placing both on the same scale, otherwise this will produce seriously distorted results.

There are other errors besides this one, and CR has an additional problem compared to MR, because for the same investment, with the same quality over time, when considering longer periods, it is observed that the MDD tends to be greater. This effect is predictable, as it is a property of the Wiener process [19]. The maximum expected amplitude in a Wiener process is proportional to the square root of time (assuming that the same strategy, same leverage, etc. are maintained).

Figure 11 demonstrates an example with ten Wiener processes.

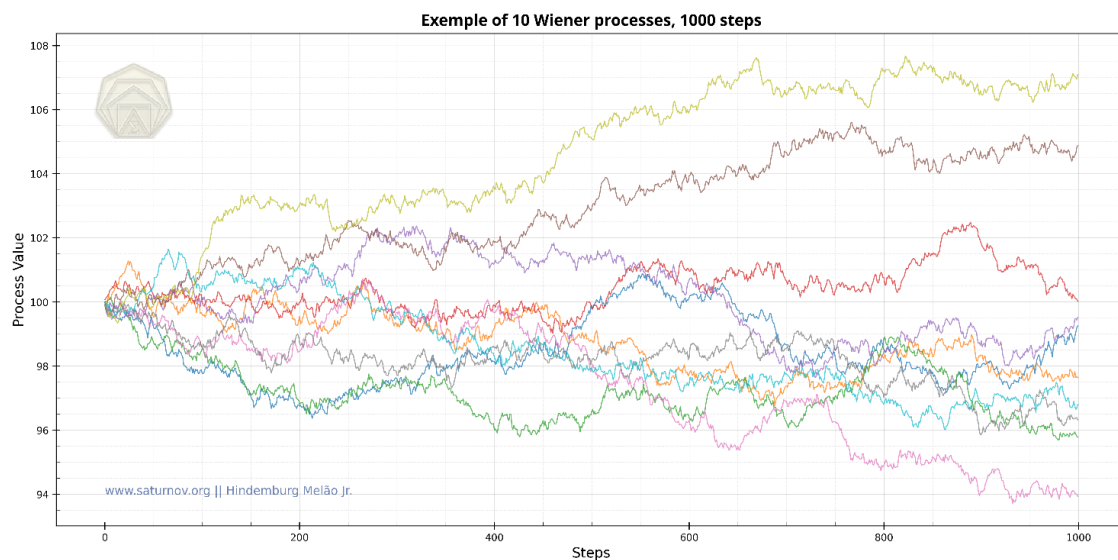


Figure 11 - Trajectories of ten Wiener processes.

This figure illustrates the evolution of ten independent Wiener processes, with one thousand increments each, represented by each colored line. The x-axis indicates the

“time” (number of “steps” or increments), while the y-axis represents the value of the process at time x .

We can see that, on average, the longer the time elapsed since the beginning, the greater the probability that the maximum produced will reach a higher value. Symmetrically, the longer the time elapsed, the greater the probability that the minimum produced will reach a lower value. In this example, the initial value is 100, and with ten processes the maximum reached close to 107 while the minimum reached close to 94, both of which occurred near the end of the curves (after 1000 increments).

If we consider the maximum and minimum so far in 200 increments, we can see that the maximum was around 103, while the minimum was close to 97. The range of variation in a 200-period interval was smaller than in a 1000-period interval. This means that if the risk is maintained over time, the MDD will increase, even if the risk remains the same.

This effect is not easy to see when considering just one growth curve, but when considering ten curves, it can be seen that, on average, across these ten curves, the worst ones are reaching lower points over longer periods, and the greater the number of periods, the lower the lowest value reached (and, symmetrically, the higher the highest value reached).

This does not happen because some Wiener processes are ‘worse’; in fact, they are not worse at all, as they are all ‘equal’ in their parameters (in their formation laws), because in this theoretical example, they all receive independent increments whose distribution of the sizes of these increments is the same in all 10 processes, therefore all these curves are equivalent. If one became better or worse than the other, it is not because it is “better” or “worse”, it is not because it used a superior strategy. It is simply the result of chance. The purpose of using several curves in the example is to have a sample of varied behaviors for the same configuration, generating several curves with the same statistical properties, and thus better visualize the effect that, on average, the MDD becomes larger over time.

In figure 12 we have one hundred Wiener processes.

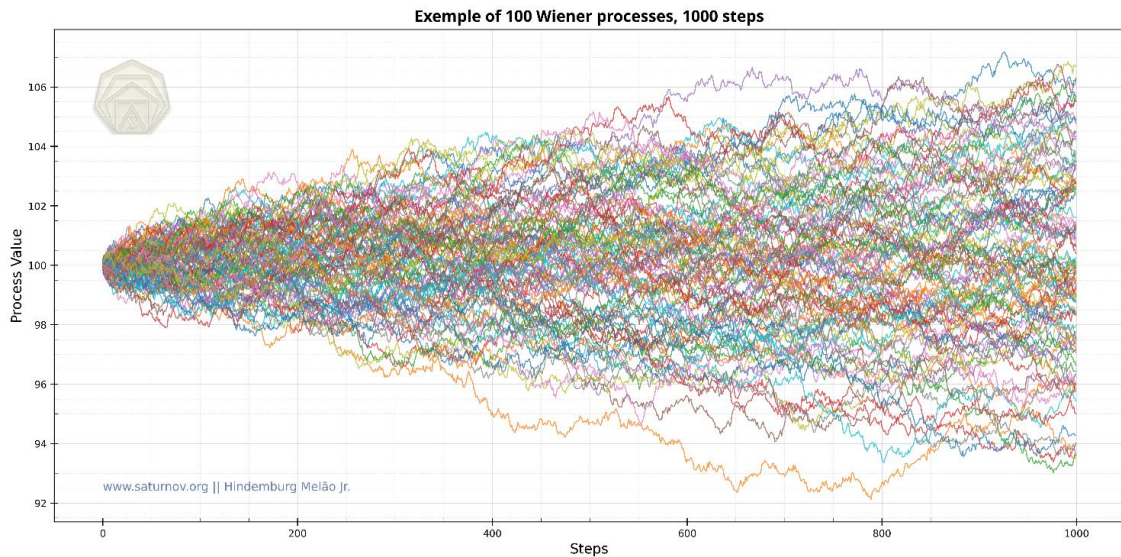


Figure 12 - Trajectories of one hundred Wiener processes.

In addition to the previous effect becoming more clearly perceptible, it also begins to take shape as an approximate “lying down” parabola, indicating that the amplitude varies with the square root of time. After 800 periods, the amplitude is about twice as large as after 200 periods, because $800/200 = 4$ e $\sqrt{4} = 2$.

With a thousand Wiener processes, the parabolic shape becomes even clearer, as shown in Figure 13.

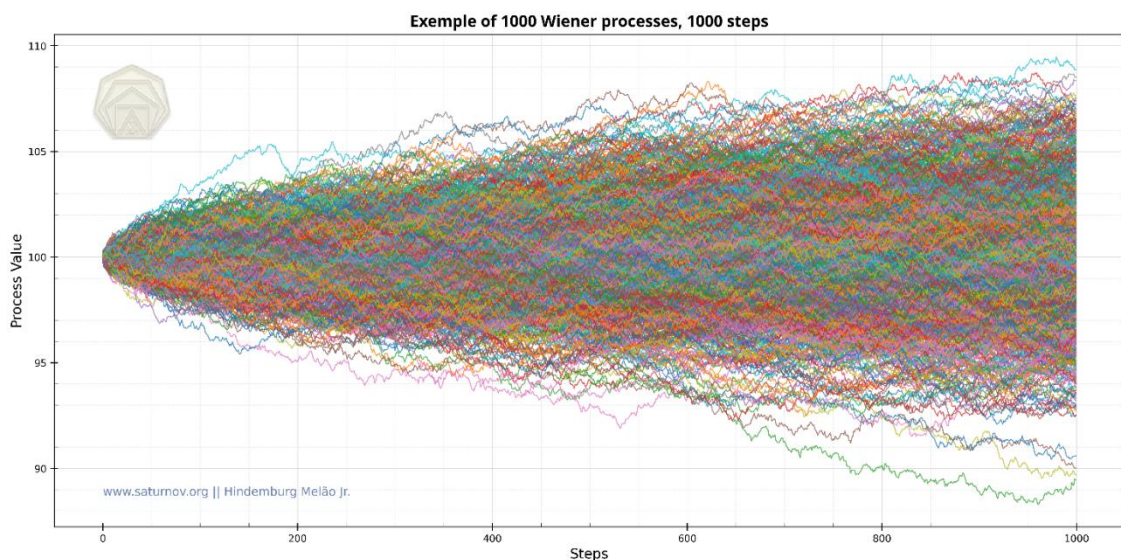


Figure 13 - Trajectories of a thousand Wiener processes.

With ten thousand and one hundred thousand Wiener processes, we can clearly see the contour in the form of a parabola in figures 14 and 15 respectively.

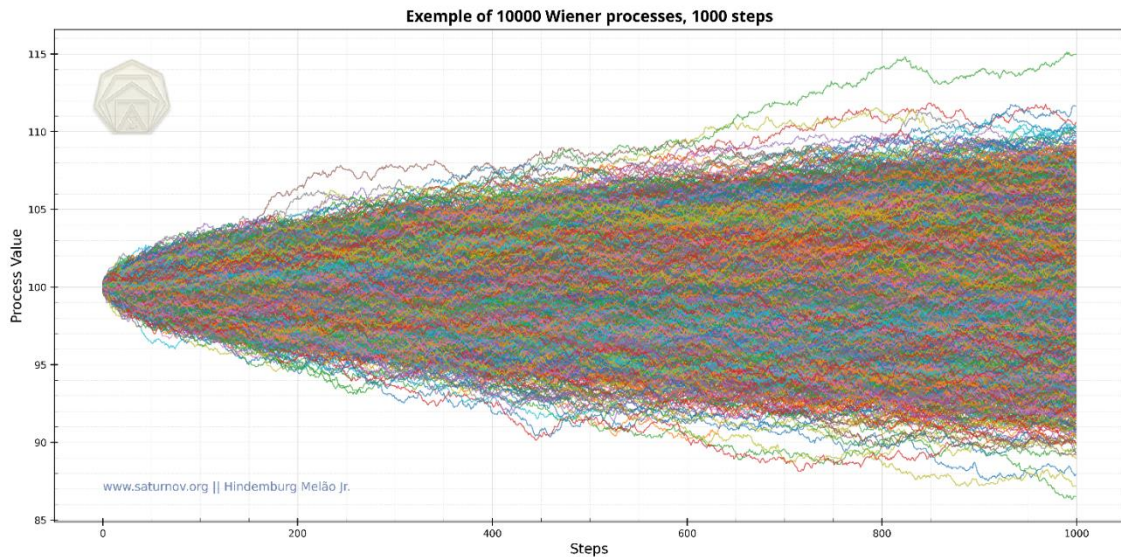


Figure 14 - Trajectories of ten thousand Wiener processes.

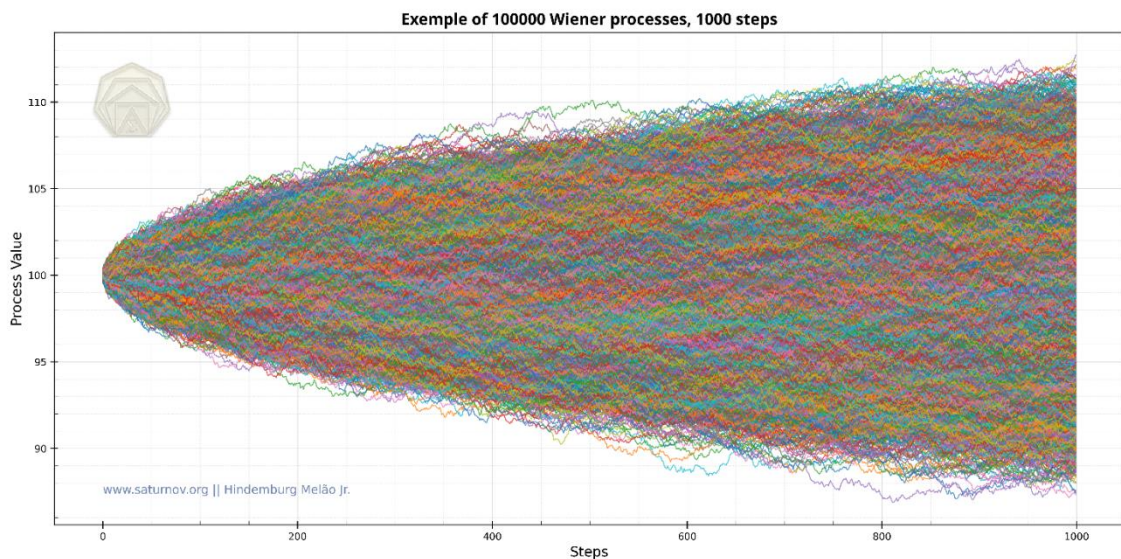


Figure 15 - Trajectories of one hundred thousand Wiener processes.

Therefore, an MDD of 50% observed over a 36-year period represents a lower risk than an MDD of 40% over a four-year period. Because over a 36-year period, since the interval is nine times greater than four years, a three-times greater MDD would be expected (because $\sqrt{9} = 3$).

Here it is necessary to make one more clarification. An MDD of 20% does not indicate a risk twice as high as an MDD of 10%, because the MDD is an asymptotic measure between 0 and 1.

If the MDD ranges between 0 and 100%, what would be twice the risk of a 60% MDD? Or twice the risk of a 90% MDD? Obviously it would not be 120% or 180%, because the

limit is 100%, you cannot have a MDD greater than 100%. Of course, leveraged operations in low liquidity markets can result in losses greater than 100%, but any MDD reaching or exceeding 100% has the same effect: total loss and inability to continue operating. Since MDD is on a scale ranging from 0 to -1, while profit is on a scale ranging from 0 to ∞ , the correct way I propose to calculate these proportions is by converting MDD to MDD* (maximum star *drawdown*) as we can see in (3):

$$MDD^* = \frac{MDD}{1-MDD} \quad (3)$$

This turns MDD into a variable MDD* that has no asymptotic limit, making it possible to perform any arithmetic operation between MDD* and CAGR without the results being distorted due to scale mismatch. So let's see how the calculation works for both 60% and 90%:

MDD 60% corresponds to $0.6 / (1-0.6) = 1.5$

$MDD^*_{(MDD=60)} = 1.5$

MDD 90% corresponds to $0.9 / (1-0.9) = 9$

$MDD^*_{(MDD=90)} = 9$

Then:

Double MDD 60 = double MDD* 1.5 = MDD* = 3

Double MDD 90 = double MDD* 9 = MDD* = 18

Now just convert back $MDD^*_{(3)}$ and $MDD^*_{(18)}$ to MDD:

$MDD_{(60\%)} = MDD^* / (1+MDD^*) = 3 / (1+3) = 3/4 = 0.75 = 75\%$

$MDD_{(90\%)} = MDD^* / (1+MDD^*) = 18 / (1+18) = 18/19 \cong 0.94737... \cong 94.74\%$

Therefore twice the 60% risk of MDD is 75% and twice the 90% risk of MDD is ~94.74%.

The graph below exemplifies this effect. For the range between 0% and 20%, it is almost impossible to notice the difference between gains and losses; the curve is almost vertically symmetrical.

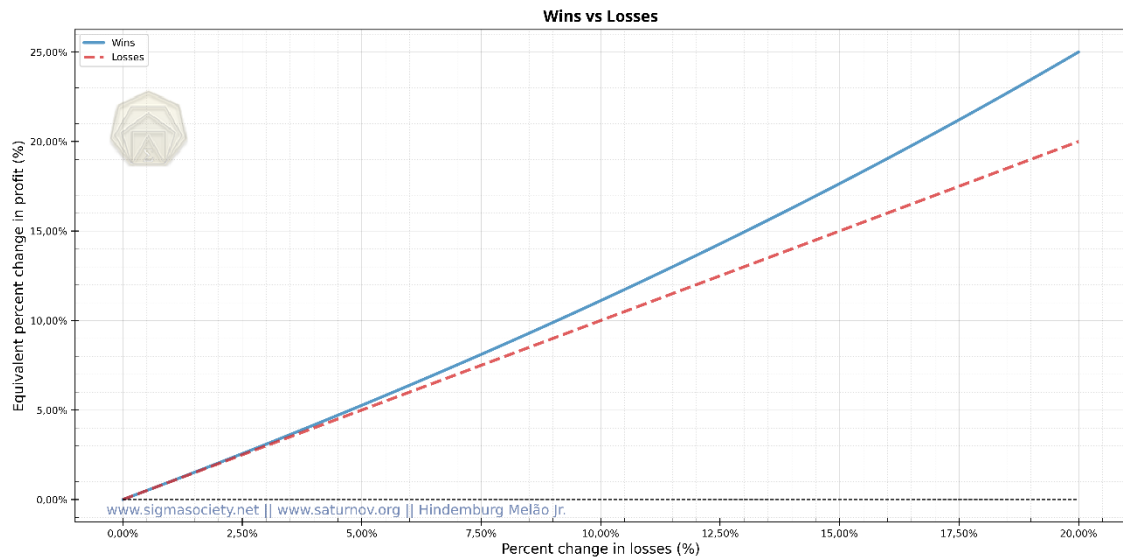


Figure 16 - Comparison of gains and losses for loss ranges from 0% to 20% (dotted red line) and equivalent gains (solid blue line).

Figure 16 shows that the gain (blue) and loss (red) curves grow at almost the same rate at the beginning, but the gain curve grows faster and faster, while the loss curve maintains a constant rate. In this range from 0 to 20% the difference is not large, but when the range from 0% to 70% is considered, the effect becomes clearer.

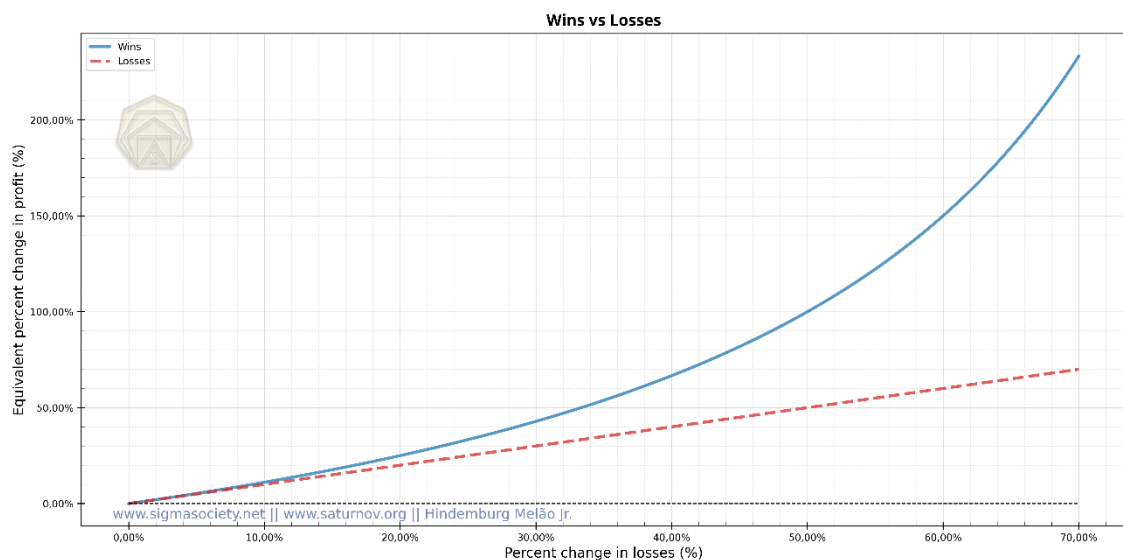


Figure 17 - Comparison of gains and losses for loss ranges from 0% to 70% (dotted red line) and equivalent gains (solid blue line).

In Figure 17, when expanding the return range to 0% to 70%, the gain curve (blue) appears well above the loss curve at the upper end, highlighting the asymmetry between the scales used to measure profitability and risk, and emphasizing the need to place both on the same scale so that the calculations do not present distortions.

When considering the range from 0% to 95%, the asymmetry becomes very large.

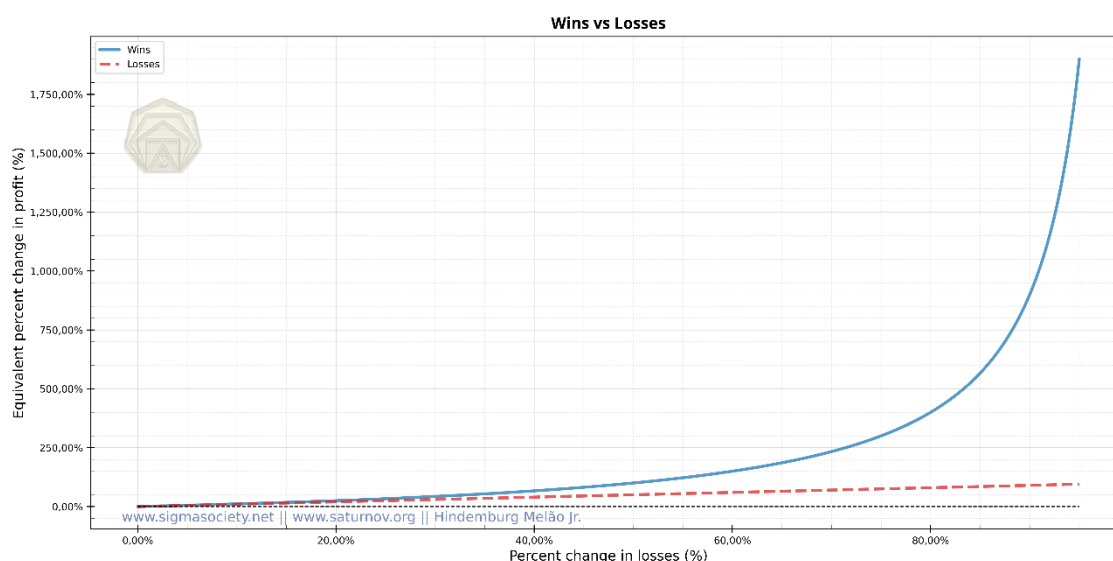


Figure 18 - Comparison of gains and losses for loss ranges from 0% to 95% (dotted red line) and gain equivalent (solid blue line).

In figure 18 the gain curve (blue) shows progressively faster growth than the loss curve (red), highlighting a large and growing asymmetry in the relationship between profitability and risk, especially at the upper end.

Figures 16, 17 and 18 illustrate the consequence of one of these scales having an asymptotic limit while the other does not. A 10% loss requires a profit slightly greater than 11.11% to be fully offset. A 20% loss requires a 25% profit to be fully offset. A 50% loss requires a 100% profit to be fully offset. A 90% loss requires a 900% profit to be fully offset. A 99% loss requires a 9900% profit to be fully offset and return to the previous value. For more information, see the percentage loss values and their respective equivalent gains in Table 1.

Percent Loss	Equivalent Gain
0,01%	0,010001%
0,10%	0,100100%
1%	1,010101%
2%	2,040816%
3%	3,0928%
4%	4,1667%
5%	5,2632%
8%	8,6957%
10%	11,1111%
15%	17,6471%
20%	25,00%
25%	33,33%
30%	42,86%
35%	53,85%
40%	66,67%
50%	100%
60%	150%
70%	233%
80%	400%
90%	900%
95%	1900%
96%	2400%
97%	3233%
98%	4900%
99%	9900%
99,90%	99900%
99,99%	999900%

Table 1 – Percentage losses and respective equivalent gains.

And so on, as the loss approaches 100%, the profit needed to recover this loss approaches infinity. Zero on one scale corresponds to zero on the other, but 1 on one scale corresponds to infinity on the other, so that the closer the value gets to 1, the larger the corresponding value on the other scale, and vice versa (the higher the value on the scale without asymptote, the closer the value on the asymptotic scale will be to 1).

In short, the greater the loss, the greater the asymmetry in determining the size of the profit needed to restore the initial value. It can also be summarized in a formula where a loss of $x\%$ requires gaining $y\%$ to return to the original value, where $y = \frac{1}{1-x}$.

This property shows that there are inconsistencies in the MR, CR, SR and other formulas that divide profitability by MDD without first converting MDD to an appropriate scale. MDD* fulfills exactly this function.

That's why the formula $CR = \frac{CAGR}{MDD}$ contains problems and distortions.

The first is that MDD is not on the same scale as CAGR. Instead of MDD, you need to use MDD*.

The second is that CAGR tends to be stable over time, while MDD tends to increase over time. Therefore, it is necessary to use an adjustment factor that considers that the

amplitude of MDD^* increases in proportion to the square root of time. If a fund has a much longer history, it is expected to have a higher MDD not because the risk is greater, but rather because for the same level of risk there is a greater probability that the MDD will be higher over a longer history, as we saw in figures 11 to 15.

The fact that MDD grows over time has led to even greater distortions in the formula, which in some sources recommend using the cumulative total return (instead of using the annualized return) divided by MDD. But the cumulative return grows exponentially over time, while MDD^* increases proportionally to the square root of time; they are very different rates. Furthermore, it is MDD^* that grows with the square root of time, not MDD itself.

The MR attempts to remedy this distortion by imposing a fixed 3-year interval, correcting this error but generating other negative effects, because if a fund has been in business for 40 years, while another fund has been in business for 3 years, it is not appropriate to disregard 37 years of historical data from the longest-running fund and compare only the last 3 years, because the history of the oldest fund is valuable information that should be used in the decision-making process. Instead of discarding everything that exceeds 3 years of history, it is necessary to estimate how much of the history implies an increase in the MDD^* and then make the comparison in a fair and well-founded manner. Of course, there are more complications to be considered, because in different scenarios, there may be a greater MDD in a few months of crisis than in many years of growth. But considering exclusively the variables in question, this is a more correct approach.

On the one hand, this is a well-known effect. On the other hand, this error in CR, MR and SR continues to perpetuate itself, without having been pointed out until now, so it may be appropriate to include this clarification.

For all that has been explained above, it is not possible to simply divide profit by MDD, as proposed in MR or CR. First, it is necessary to normalize the scale. To do so, one can choose between two equivalent procedures:

1. Use MDD^* instead of MDD, then divide the profitability by MDD^* .
2. Transform the profitability into an asymptotic scale, with the same properties as the MDD, and then divide by the MDD.

The procedure described in 1 is preferable because it places both variables on a proportional scale, facilitating all subsequent operations.

It would not be wrong to use the procedure in 2, but all equations would need to be adjusted according to the properties of the scale with asymptotic limits, which would make all steps more laborious and less intuitive.

It is worth noting that, strictly speaking, it is not possible to fully normalize the scales. This happens because it is not possible to reinvest continuously, that is, the model treats the variables as if they were continuous, but in fact they are discretized in cents, *pipets*, or points, and this has some consequences.

Therefore, although MR, CR, and SR are superior to ShI and SoI in determining the true maximum risk, these metrics have other flaws and continue to be unable to satisfactorily deal with funds with different longevities, in addition to generating distorted scores when dividing the CAGR by MDD, without first normalizing these variables.

There are other problems, such as the need to use logarithms of the values instead of using the values themselves, the need to subtract inflation and standardize profitability (annual, for example) instead of considering the entire accumulated profit. There are other problems with MR and CR, so the best decision may be to create a consistent and theoretically sound metric from scratch.

But this is still far from resolving all the distortions in these indicators, and even if all the normalization problems were resolved, there would still be the probabilistic issue of very large uncertainty in the measurement, because the MDD is an isolated element, it is not a data sample, so the uncertainty in this value is large compared to the volatility measured in the ShI or SoI.

In this regard, one might be under the illusion that it would be more advantageous to use ShI or SoI, because both are based on samples with several measurements, and from these the risk is estimated. The problem is that the risks calculated by these indexes are underestimated. The precision in ShI and SoI is greater, but the accuracy is lower. Therefore, the use of MDD is preferable.

The empirically measured MDD shows that that was the MDD in that period, while the ShI or SoI try to estimate what the MDD would be based on the mean and standard deviation of the oscillations, starting from a false hypothesis (that the distribution of the oscillations is Gaussian), or worse, they do not try to estimate the maximum risk, they just use the mean risk (the standard deviation) as a parameter.

It is also important to keep in mind that using the average of the MDDs does not solve the problem. In fact, it makes the problem worse, because obviously the second MDD is smaller than the first, and the third is smaller than the second, and so on, so the average of the MDDs will not indicate the maximum risk better than the MDD itself. What one can do is try to use the largest MDDs recorded to project the MDD over a given time horizon, together with the expected MDD based on the widening of the oscillation amplitude in a Wiener process as a function of time, using a Bayesian approach.

There are different ways to deal with this. One way is to use the second MDD to estimate the global MDD from it, and then take the average of the measured MDD and the MDD estimated from the second MDD. To do this estimate, simply find the *ranking positions* of each MDD and use those positions to estimate what the depth of the global MDD would be.

For example, in a 100-month history we have MDD 38.0% and second MDD 34.5%. If the balance sheet measures are updated monthly, then MDD 38.0% is at the 99th percentile and MDD 34.5% is at the 98th percentile. Assuming the distribution is normal (usually it is not, but in this first example, let's assume it is), then 38.0% corresponds to 2.33 standard deviations above the mean and 34.5% corresponds to 2.05 standard deviations above the mean.

Let's say the monthly average is 6.5%. Then we can see that the standard deviation value calculated based on the largest MDD is approximately 13.54% and based on the second largest MDD it is 13.63%. To estimate what the MDD would be based on the measurement of the second MDD, simply do $2.33 \times 13.63\% + 6.5\% \cong 38.26\%$. The basic idea is to calculate the standard deviation for the second MDD and use this standard deviation to determine what the MDD would be if the sample dispersion were the one estimated based on the position of the second MDD.

The same can be repeated for the third MDD and so on. Of course, this is much more appropriate than directly averaging the MDD, the second MDD... the nth MDD, since direct averaging would not provide an adequate estimate of the maximum risk.

Once you have obtained the different MDD estimates from the k-th MDDs, you do not need to average them all. Another way to use these results is to consider the largest of them (smallest if you consider them to be negative). This may be a better estimate of the maximum risk over the period.

The correct calculation would actually require first converting each MDD to an MDD* and following the other procedures outlined above. The important thing here is just to understand how one can use the second MDD, the third MDD, etc. to make estimates of MDD values, to generate a realistic empirical sample of several MDDs, and then obtain an average whose uncertainty is smaller than the uncertainty in the MDD itself.

It is also important to keep in mind that since these distributions tend to have dense tails, it may be advisable to apply an attenuation factor in the calculation to refine the results. Alternatively, one can test different distributions to investigate which one best fits the empirical data and use that in the calculations, instead of adopting a Gaussian.

The rate at which the MMD grows over time can also be modeled using increments whose distribution is not normal, instead of using traditional Wiener processes. But this would require knowledge of data that are generally not available for the vast majority of funds. This would be useful for automated investment systems that can be backtested thousands or millions of times, and thus know the true distribution of the curves for a variety of genotypes. But for funds with human managers this alternative is less viable, and it is usually necessary to assume the normal distribution of the increments.

There are several other details that need to be considered here. To begin with, it would be necessary to establish a rigorous definition of “second MDD”. In the example above, we treated monthly MDD in the balance sheets, because it is more didactic, but the true MDD needs to be accounted for in the *equity* at each *tick*, and when considering *tick-by-tick* we have the problem that the second MDD is obviously the *tick* immediately after or immediately before the MDD, and this criterion would be useless for the calculation.

One could consider MDD only those events after it returns above the high-water mark after the previous MDD. This would reduce the sample to a few units, but would have the advantage that each MDD would be conceptually quite useful for the purpose of estimating the overall MDD.

It should also be noted that portfolio *equity* is not exactly a Wiener process, although it can be treated as if it were (it is a good approximation). Therefore, one cannot indiscriminately use valid hypotheses for a Wiener process as if they were equally valid for a historical price or the historical fund. They must be used, but within the limits that are appropriately applicable.

VIII. NEW METHODS FOR MEASURING INVESTMENT QUALITY: *MELAO INDEX AND VARIANTS*

Considering all the aforementioned aspects, a consistent method for measuring investment quality must, at a minimum, take into account profitability and risk, placed on an appropriate scale, measured appropriately, respecting the properties of the variables and scales involved, to avoid distortions or at least minimize them.

The MeI, which will be proposed in different versions, allows for measuring investment quality consistently while reducing distortions observed in other indexes. There is no way to completely eliminate all distortions, because some are associated with intrinsic properties of the Financial Market, such as the discretization of quotations, for example.

The primary version of the MeI is as follows, as defined by equation (4):

$$Melao_{Index} = \frac{\ln(1+R) - \ln(1+i)}{\ln(1+MDD^*)} \sqrt{T} \quad (4)$$

where R is the average annual return on investment, i is inflation, and T is time.

The MDD* can be either the MDD* itself, calculated from the MDD, or the average of the MDD* estimates based on the k-th MDD*, or even the best of the options which is the maximum among the MDD* estimated from the k-th MDD*, as they are the best representation for the true maximum risk measured within the considered interval.

This version incorporates a synthesis of all the corrections analyzed so far, and avoids all the distortions pointed out in other indexes. This does not mean that it is a perfect or complete solution, but it offers important advantages compared to any other existing index.

The main advantages of the MeI over ShI and SoI consist in applying the same correction as MR or CR, i.e., replacing the standard deviation with MDD, but without the distortions present in MR and CR. This increases safety by increasing sensitivity to *outliers*. In addition, it corrects all scale distortions present in ShI, SoI, MR, CR, MRAP and others.

As an alternative to MeI, the corrected versions *MShI* and *MSoI* make it possible to use the standard deviation as a measure of risk, but normalizing the scales in order to eliminate the errors present in ShI and SoI, in addition to neutralizing the distortions caused by inflation, resulting in more appropriate metrics.

The distortions produced by using formulas that involve values that are on different scales, without first standardizing the scale, are equivalent to calculating the average body temperature of people in a room, using measurements in degrees Celsius for some people and degrees Fahrenheit for others, adding everything up and dividing by the number of people. This should never be done. Before performing any arithmetic operation with the values, it is necessary to place all the temperatures on the same scale (preferably in Kelvin, if the procedure involves operations other than addition and subtraction).

The effect produced by inflation can be compared to people inside a large ship. The ship goes from one country to another, and people walk inside the ship. There is usually no correlation between the trajectories of people inside the ship most of the time, but since they are all being carried by the ship, if you measure the positions of these people in relation to the surface of the Earth, you get the illusion that they are all moving almost together. However, if you measure their trajectories in relation to the surface of the ship, you realize that they are moving almost randomly, and may occasionally move together at specific times, such as during meals or when they go to bed, or at some event on the ship that brings them together, but most of the time they move almost independently of each other. In this analogy, the index, such as the S&P 500 or IBOV, is equivalent to the ship, which drags the performances of the funds along with it. If the IBOV goes up, the funds profit, and vice versa, regardless of any strategy. In this situation, if last month the ship's movement followed approximately the same direction as the month before, one is left with the sensation of homogeneity in people's directional behavior over time, when, in fact, individual movements are quite different from each other, and in order for them to be correctly studied and understood, it is necessary to subtract the ship's movement.

Therefore, when analyzing the behavior of funds, it is necessary to subtract the movement of the "current", caused by the stock market index, before evaluating the quality of the investment, because when judging the quality of a fund, we are not interested in the movement of the IBOV or in how much the fund is "dragged" by the IBOV, but rather in the fund's "own" movement, that is, the movement of the fund in relation to the IBOV, or in relation to the S&P 500, or in relation to the markets whose assets are components of the portfolio of that fund. Of course, this applies to funds that carry out long-term operations.

The ShI attempts to subtract "something", but the way it has been done is not correct. Usually, a "risk-free *benchmark*" is chosen, but there are three errors in this choice. The

first error is that there is no “risk-free benchmark”. The correct thing to do is to subtract two distortion factors:

1. Inflation, to place all quotations on an inertial reference scale as close to zero as possible.
2. The stock exchange index (or Financial Market) in which operations are carried out, to neutralize gains and losses caused by the “current”.

These two factors point to the other two errors. Sharpe’s idea of subtracting a “risk-free benchmark” is not entirely wrong, but it is not the most correct way to deal with this problem.

IX. CONCLUSION

There are four important points to highlight:

1. The way performance has been measured so far is adequate for accounting purposes and financial statements but is not suitable from a mathematical and statistical perspective. It fails to properly assess investment quality as it is heavily dependent on the choice of starting and ending points. This article presents a simple and effective solution to this issue.
2. The indexes used to evaluate risk-adjusted performance are based on formulas that present conceptual and quantitative issues, sometimes involving divisions between variables on different scales, which leads to distortions and, consequently, errors in selecting the best investments. This article introduces the necessary corrections and improvements.
3. Additional distortions and conceptual errors in these formulas are addressed, particularly those involving proportions between variables that grow in geometric progression (GP) but have been treated as if they followed an arithmetic progression (AP). This misinterpretation results in further distortions that hinder the selection of the best investments.
4. In some cases, correcting the distortions described in items 2 and 3 is not worthwhile because the structure of the formula itself is inherently flawed, relying on naive assumptions about the distribution of fluctuations. To address this issue,

we propose a new index (MeI), which does not claim to be a definitive solution but represents significant advancements over existing alternatives.

The MeI incorporates fundamental corrections and improvements for the accurate comparison of investment funds and introduces more appropriate criteria for measuring performance. As a result, it provides a more robust, reliable, and precise index for evaluating investment quality. The MShI and MSoI further complement this review and enhancement process by addressing cases where the standard deviation – or another measure of dispersion – is preferable to MDD.

With these advancements, we aim to provide sophisticated investors with enhanced tools to assess and select the most profitable and secure investment alternatives. This is achieved by refining traditional methodologies and introducing innovative approaches, ultimately offering more precise, effective, and reliable techniques for investment evaluation.

APPENDIX A - DIFFERENCE IN THE USE OF “(INVESTMENT-INFLATION)- (RISK-FREE INVESTMENT-INFLATION)” AND “INVESTMENT-RISK-FREE INVESTMENT”

When you include inflation in both cases, you are considering the volatility in inflation, and the accumulated history of fluctuations in “investment-inflation” can be very different from the history of fluctuations in “investment”, and the same when you consider “risk-free investment-inflation” and “risk-free investment”. So although $(A-B)-(C-B)$ can be simplified to $A-C$, this doesn't apply when it comes to a historical series in which the evolution of the values of $(A-B)$ follows a very different course from that followed by A , and the same goes for $(C-B)$ and for C . Since volatility is an element in these calculations, it is necessary to consider the evolution of fluctuations in each case.

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